STONE

# Intelligent TFT-LCD Module

# Model
# STI080WT/N-01

**Equipment Manual**

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# Preface

This equipment manual is part of our Intelligent TFT-LCD Module documentation. It provides the information in regards of operation, installation, configuration, function, system as well as its technical design and working principle.

### Organization of the manual

The STI080WT/N-01 equipment manual is organized into the following chapters:

| Chapter | Contents |
|---|---|
| **1** | Overview of features and functional scope of the STI080WT/N-01 |
| **2-3** | Technical Parameters, Interface Description |
| **4-6** | Accessories, Installation, Physical Dimensions |
| **7** | Command Set |
| **8-10** | Electrical Components, Naming Rule, International Certification |
| **Appendix** | MCU Sample Program, MCU Circuit Design, ESD Guidelines |

### Intelligent Customer Online Services

Intelligent Customer Support offers comprehensive additional information of Intelligent Products through its Online services as follows:

- Official website: http://en.stone-hmi.com/
- Alibaba platform: http://shuangying-ocean.en.alibaba.com/
- Or through telephone 0086-10-84351669

### Other support

In need of technical queries, please contact STONE representatives in the subsidiaries and branches responsible for your area.

## Trademarks

STONE registered trademarks are as below:

- STONE
- STONE TECH
- Intelligent HMI
- Intelligent TFT-LCD Module

## Abbreviations

The abbreviation table in this equipment manual is as below:

| | |
|---|---|
| LED | Light Emitting Diode |
| CPU | Central Processing Unit |
| ESD | Electrostatic Sensitive Device |
| HMI | Human Machine Interface |
| IF | Interface |
| LCD | Liquid Crystal Display |
| UART | Universal Asynchronous Receiver/Transmitter |
| COM | Commercial |
| DIN | Data Input |
| DOUT | Data Output |
| VIN | Voltage Input |
| GND | Ground |
| TP | Touch Panel |

A list of all the technical terms together with their explanations is provided in the glossary at the end of this manual.

# 1 Introduction

**This chapter contains general information of:**

- Brief Introduction
- Warranty
- Product Characteristics
- Application Area
- Working principle
- Operation Processing
- Software Operation

## 1.1 Brief Introduction

The STI080WT/N-01 has been conceived as **TFT monitor** & **Touch controller.** It includes processor, control program, driver, flash memory, RS232/ RS485 port, touchscreen, power supply etc., so it is a whole display system based on the powerful & easy Command Set operating system, which can be controlled by Any MCU.

The STI080WT/N-01 can be used to perform all basic functions, such as text display, image display, curve display as well as touch function etc. The User Interface can be more abundant and various. And the flash memory can store your data, configuration files and images etc.

## 1.2 Warranty

All products purchased from our company are guaranteed to keep in good repair for 3 year s. If quality problems (except human error) happen in guarantee period, our company will maintain for free or replace the broken one unconditionally.

## 1.3 Product Characteristics

- With CPU & Driving device
- Controlled by any MCU
- Display Picture/ Text /Curve
- 65536 colour TFT display
- With/without Touch Screen
- RS232/ RS485/ TTL UART Interface & USB port
- Wide voltage range
- Easy to use! Powerful function! Saving cost and time!

## 1.4 Application Area

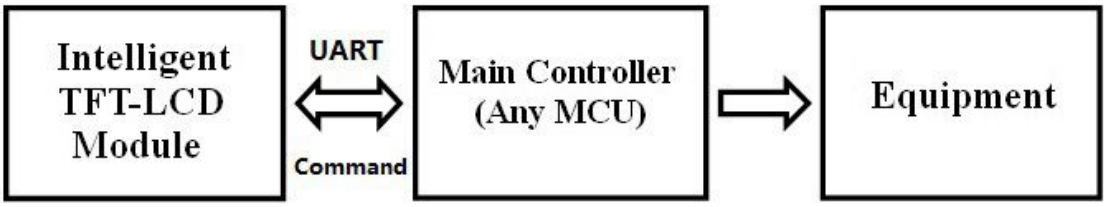
Widely used in various industrial field

- Medical & Beauty Equipment
- Engineering Machinery and Vehicle Equipment
- Electronic Instrument
- Industrial Control System
- Electric Power Industry
- Civil Electronic Equipment
- Automation Equipment
- Traffic

Etc.

### 1.5 Working Principle

The Intelligent TFT-LCD Module communicates with the Customer's MCU via Command Set (HEX Code), and then the MCU would control its connected equipment to work according to the received commands.



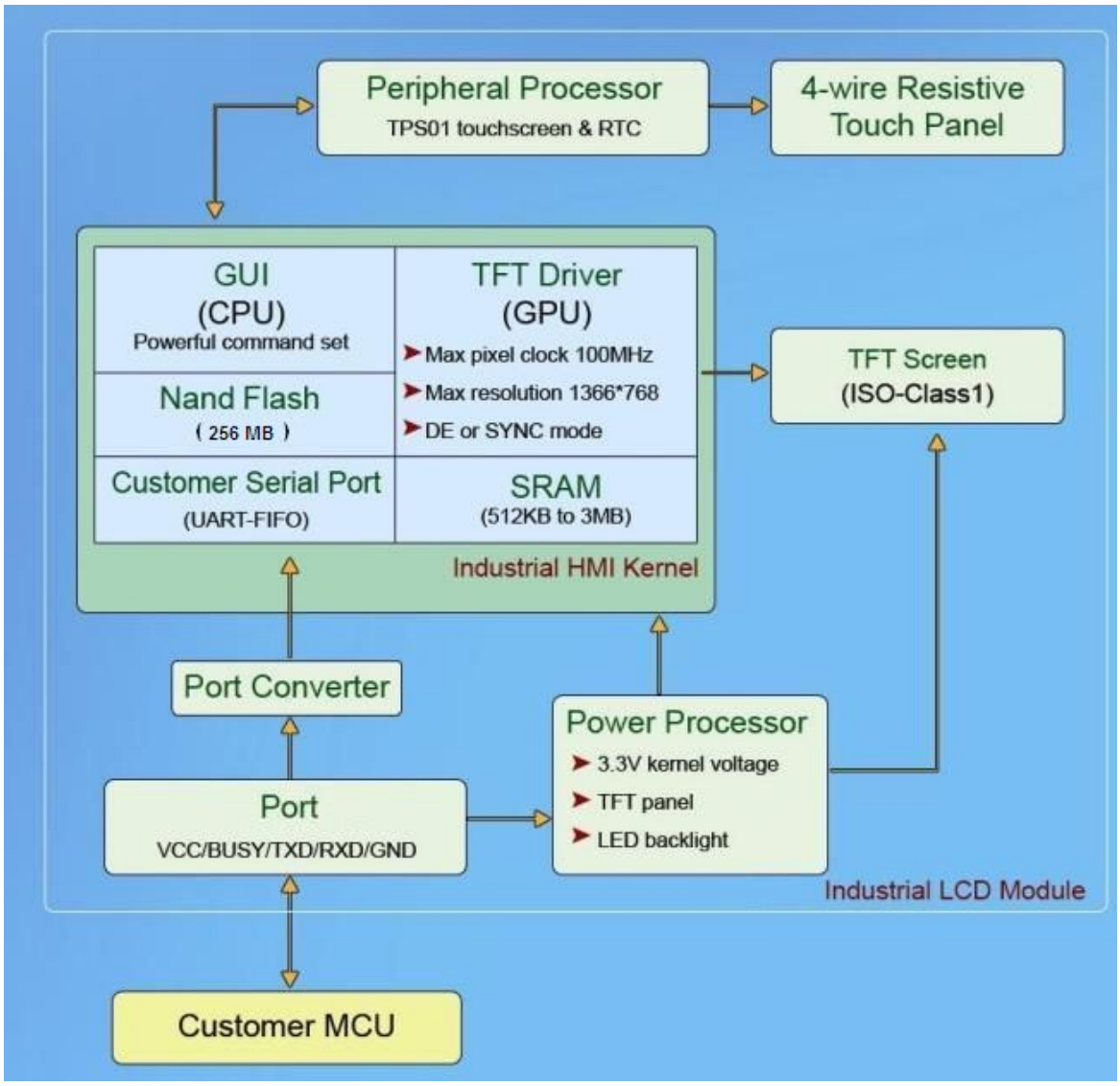




Figure 1.3-1 Configuration and process control phases

## 1.6 Operation Processing

Only **3 steps** to use our TFT-LCD Module:

1. Design a group of Beautiful “Graphical User Interface”. (Ref. Picture 1.4-1)
2. Connect with customer’s MCU through RS232, RS485 or TTL level directly. Plug and play.
3. Write a simple program for MCU to control the TFT-LCD Module via powerful “Command Set” (HEX Code).

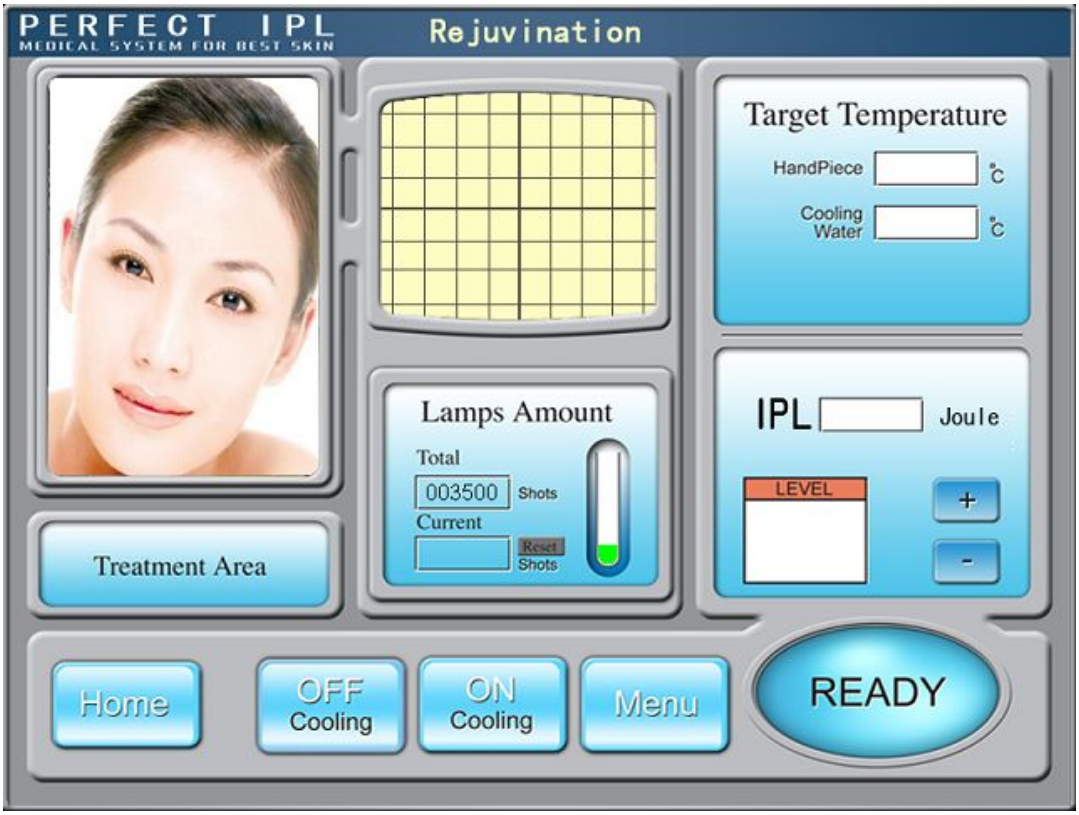
That’s all.

For example: Image Command 0x70.

**Send: “0xAA + 0x70 + Pic_ID + 0xCC 0x33 0xC3 0x3C”**

| | Introduction | Example | Note |
|---|---|---|---|
| **AA** | **Frame head** | | **Frame head of each Command** |
| **0x70** | **Command Code** | | |
| **Pic_ID** | **Sequence number of the storage position.** | **0x00 0x01** | **Show Picture – 01** |
| **CC 33 C3 3C** | **Frame end** | | **Frame end of each Command** |

**More information, please reference the document of Command Set.**




Picture 1.4-1

## 1.7 Software Operation

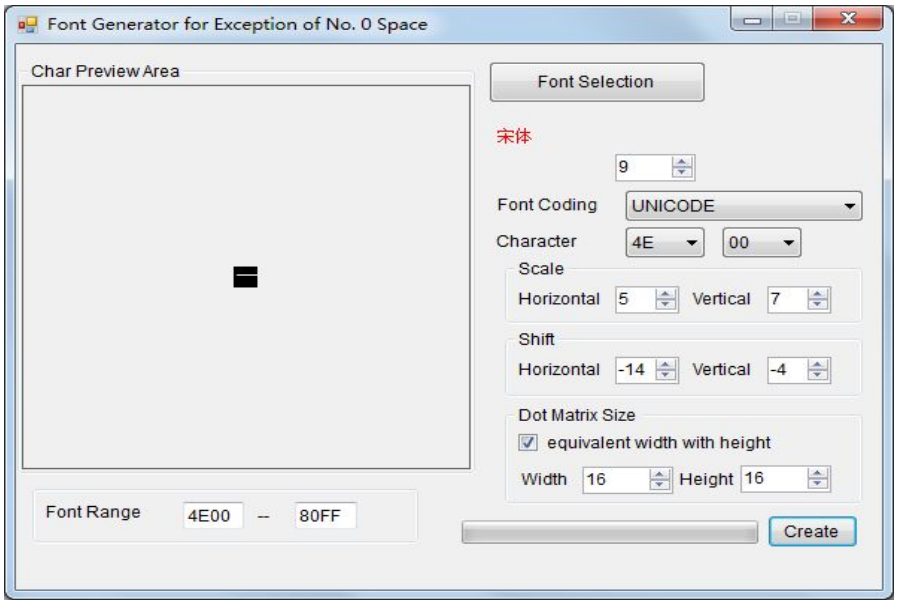
We will offer simple software “Toolbox” to help you to work with Intelligent TFT-LCD Module on computer.

1. Development Assistant: Show how to send Commands to control the TFT-LCD Module.
2. Download Assistant: Download the images and configuration files into TFT-LCD Module.
3. Update firmware: For updating firmware.
4. Touch control: Set many buttons on the User Interface, and create a touch configuration file.
5. Dial control: Create the dial configuration file.




Picture 1.5-1

6. Font software: Create font configuration file.




Picture 1.5-2

# 2 Technical Parameters

**This chapter contains technical data on:**

- **Physical Parameters:**
  - Physical Parameters
  - Display

- **Hardware Parameters:**
  - Processor
  - Memory
  - Interface
  - Power Supply

- **Storage & Test**
  - Electrical Characteristics
  - Ambient Conditions
  - Noise Immunity
  - Radio Interference

- **Support Device**
  - Support Device

| **Physical Parameter** | |
|---|---|
| Size | 8 inch |
| Resolution | 800×RGB×600 |
| Pixel Spacing | 0.1905 mm × 0.0635 mm (H×V) |
| Color | 65536 colors (16 bit) |
| Viewing Area | 162.0 mm × 121.5 mm |
| Display Dimension | 164.0 mm × 123.5 mm |
| Overall Dimension | 200.6 mm × 142.4 mm × 17.4 mm(N)/<br>19 mm(T) |
| Net Weight | 435 g(N)/ 550 g(T) |

| **Display** | |
|---|---|
| Backlight Type | LED |
| Brightness | 500 cd/$m^2$<br>(Brightness can be adjustable in 100 levels) |
| Contrast | 500:1 |
| Backlight life | 20,000 hours |
| Viewing Angle | 70°/ 70°/ 50°/ 70°(L/ R/ U/ D) |
| TFT Panel | A Class Industry Panel |
| Touch Screen | Industry Level 4 wire resistance<br>Or without touch screen is optional. |
| Screen Mode： | Digital |

| **Processor** | |
|---|---|
| CPU | CortexM3 STM32F103 70MHz |
| LCD Controller | CPLD EPM240 |
| Refresh Rate | 60 Hz |
| Update Speed of per frame | 47 ms/frame (21 images/s) |

| **Memory** | |
|---|---|
| Flash Memory | Standard 256MB, Extension 2GB |
| Memory Amount for picture | 227 images, Extension 2019 |

| **Interface** | |
|---|---|
| Interface | RS232/ RS485/ TTL UART Interface |
| Image downloading | MiniUSB2.0 (12Mbps) & SD card |

| **Power Supply** | |
|---|---|
| Rated voltage | +12 V DC |
| Permissible voltage range | +4.5~+20.0 V DC (or Extend to +42V) |
| Max. permissible transients | +24 V (500 msec) |
| Time between two transients | 50 sec minimum |
| Fuse, internal | Electronic |
| Power consumption | 4 W |

**Electrical Characteristics**

| **Parameter** | | **Condition** | **Min** | **Type** | **Max** |
|---|---|---|---|---|---|
| Supply Current | | VIN=5V | | 560mA | |
| | | VIN=12V | | 230mA | |
| | | VIN=20V | | 140mA | |
| Signal Input Voltage | TTL level | VIH | 2.1V | | |
| | | VIL | | | 0.9V |
| | RS232 level | V range | -15V | | +15V |
| | RS485 level | Different Threshold | -0.2V | | +0.2V |
| Signal Output Voltage | TTL level | VOH | 3V | | 3.3V |
| | | VOL | 0V | | 0.1V |
| | RS232 level | V range | -15V | | +15V |
| | RS485 level | Different Driver | | | 5V |
| Baud Rate | | | 1200 bps | | 921600 bps |
| Note: As the brightness lower, the current will also reduce. | | | | | |

| **Ambient Conditions** | |
|---|---|
| **Max. permissible ambient temperature** | |
| Operation | -20℃ ～ +85℃ |
| Storage | -30℃ ～ +85℃ |
| **Relative humidity** | |
| Operation | 55℃, 85% |
| Storage | 60℃, 90% |
| **Shock loading** | |
| Operation | 15 g/ 11 msec |
| Storage | 25 g/ 6 msec |
| **Vibration** | |
| Operation | 0.035 mm (10 - 58 Hz)/ 1 g (58 - 500 Hz) |
| Storage | 3.5 mm (5 - 8,5 Hz)/ 1 g (8.5 - 500 Hz) |
| **Barometric pressure** | |
| Operation | 706 to 1030 hPa |
| Storage | 581 to 1030 hPa |

| **Noise Immunity** | |
|---|---|
| Static discharge<br>(contact discharge/air discharge) | EN 61000-4-2<br>6 kV/ 8 kV |
| RF irradiation | EN 61000-4-3<br>10 V/m, 80% AM<br>1 kHz |
| Pulse modulation | ENV 50204<br>900 MHz ±5 MHz<br>10 V/meff., 50% ED, 200 Hz |
| RF conduction | EN 61000-4-6<br>150 kHz - 80 MHz<br>10 V, 80% AM, 1 kHz |
| Burst interference<br>Supply lines<br>Process data lines<br>Signal lines | EN 61000-4-4<br>2kV<br>2kV<br>1kV |

| **Radio Interference** | |
|---|---|
| Radio interference level complying to EN 55011 | Class A |

| **Support Device** | |
|---|---|
| Buzzer | Support |
| RTC | Support |
| Key-board Interface | Support<br>User can set dot matrix for 8*8 or 4*4 |
| 16 IO Ports for Users | Support |
| USB port For Downloading Pictures & Font File | Support<br>(Download one 800×600 picture only take 0.5S) |
| Touch Screen | 4 Wire Resistance |
| Default Font | 8×8 / 16×16/ 16×32/ 12×12 / 12×24<br>(Dot Matrix) |
| Picture | Support BMP, JPG Format |
| Storage Data | Support |
| Command Set | Unified Simplified Command Sets |

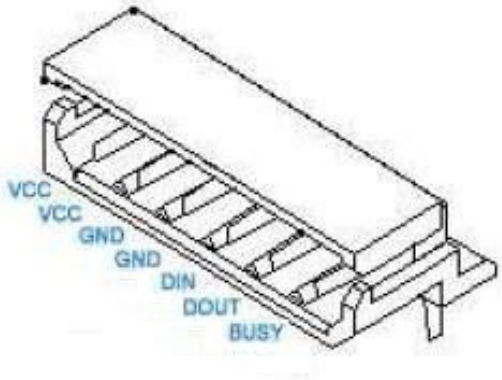
# 3 Interface Description

**This chapter contains the description of the interfaces:**

- VVC
- GND
- DIN
- DOUT
- BUSY
- Baud Rate

Please notify the interface type before ordering. RS232/ RS485/ TTL level interface.

## Communication Interface Definition：




| Pin Name | Pin NO. | Pin Type | Interpret |
|---|---|---|---|
| **VCC** | 1,2 | P | Power Supply Input |
| **GND** | 3,4 | P | Power Ground |
| **DIN** | 5, | I | Data Input |
| **DOUT** | 6 | O | Data Output |
| **BUSY** | 7 | O | Issue a directive as Full Signal of UART Buffer |

**I: Input O: Output P: Power**

**Note A**: 1. Adopting the 7 Pin 2mm spacing socket. (Socket Model: Molex 5023520700)
2. Direction of the signal was defined with TFT-LCD Module;
"I" refers to the signal from the user's system transmitted to the TFT-LCD Module.
3. Pins with the same definition are connected together in the module inside.
4. User can set the RS232 or 3.3 V TTL/ CMOS level by soldering pad on the PCB. Default is "OFF" (RS232 Level). If change to "ON" (TTL Level), only need to short the soldering pad of J21, J22, J23 and take out R61, R62, R63.
5. The RS485 and other interface can be customized which need to point out in the order.
6."Busy" signal doesn't need to be considered as communication, Frame and Command never be lost, (70MHz CPU, 1M bit Command Buffer), "BUSY" signal can be accessed to data input pin or left vacant.

**Note B**: The selection of Baud rate for the serial interface：

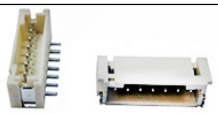
| Baud rate(bps) | 1200 | 2400 | 4800 | 9600 | 19200 | 38600 | 57600 | 115200 |
|---|---|---|---|---|---|---|---|---|
| Bode_Set | 0x00 | 0x01 | 0x02 | 0x03 | 0x04 | 0x05 | 0x06 | 0x07 |

# 4 Accessories

**This chapter contains the accessories:**

- Double 7-pin Connect Cable
- DB9 Connecting Cable
- 7-pin Socket
- MiniUSB Cable
- MCU board for DEMO
- Converter
- Bezel

*Accessories*

| Accessory Name | Model | Note | Picture |
|---|---|---|---|
| Double 7-pin Cable | L7 | Plug Model:<br>Molex 0050375083<br>Optional：<br>10cm/20cm/30cm/65cm |  |
| DB9 Cable | LD | Connector：<br>Standard DB9 Joint |  |
| 7-pin Socket | S7 | Socket Model:<br>Molex 5023520700<br>Socket Spacing : 2mm |  |
| MiniUSB Cable | LU | |  |
| MCU Demo Board | MB | Demonstrate Application |  |
| Converter | UR2.0<br>UR4.0<br>UR1.0 | USB to RS232<br>USB to RS485<br>USB to TTL |  |
| Plastic Bezel | PS-043<br>PS-050<br>PS-056<br>PS-070<br>PS-080<br>PS-104 | For: 4.3’’, 5’’, 5.6’’, 7’’, 8’’, 10.4’’ TFT-LCD Module. |  |

# 5 Installation

**This chapter contains the installation of plastic bezel.**

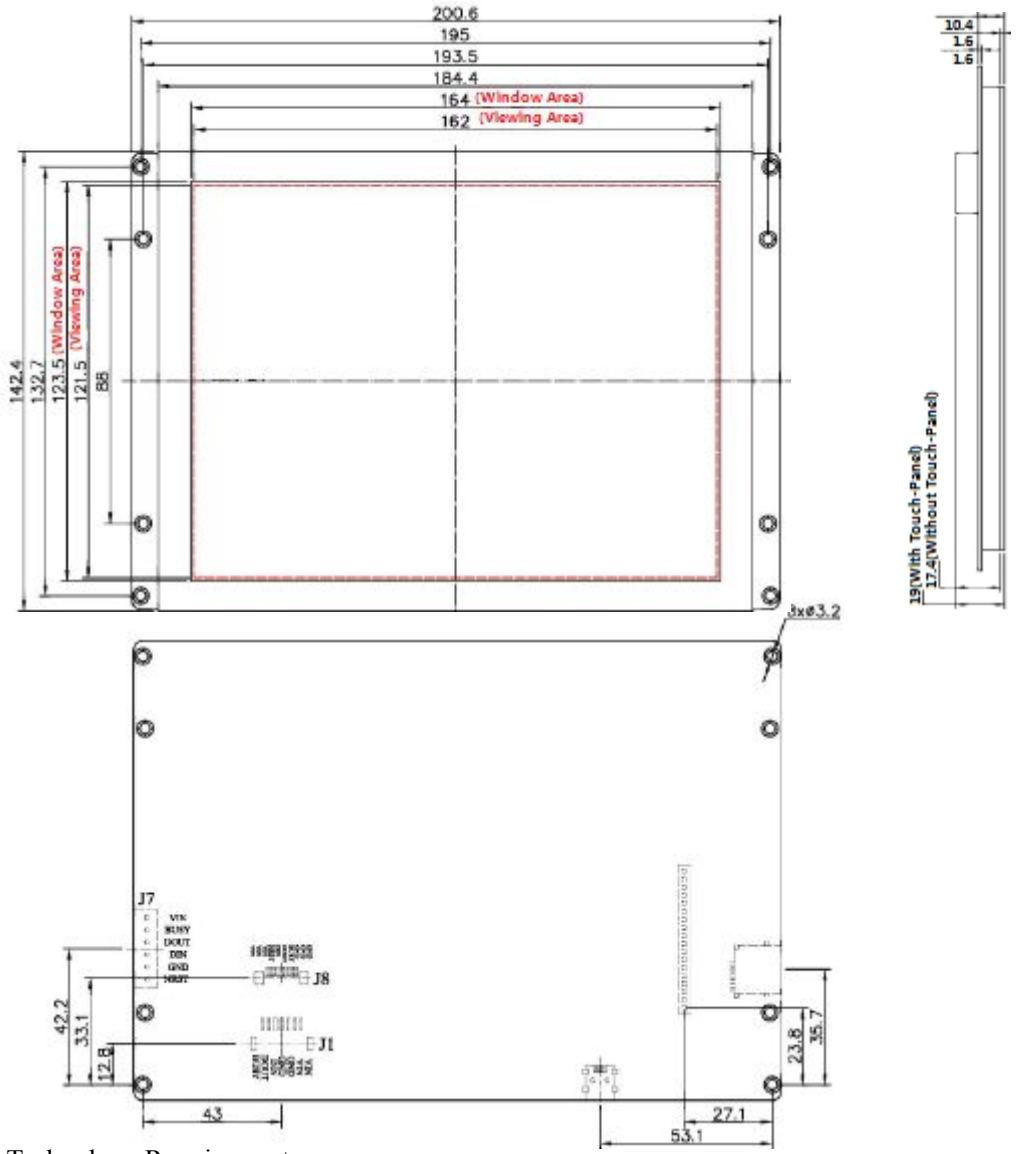


# 6 Physical Dimensions

**This chapter contains the information of Physical Dimensions.**

## Model: STI080WT-01

(Unit:mm)




Technology Requirement:

1. Display Center is used as position reference.
2. Unmark Tolerance is +/- 0.3mm.

Note: The Viewing Area is marked in red dashed.

Figure 7-1 STI080WT/N-01 dimension

# 7 Command Set Table

**This chapter describes the Command Set:**

- Text Command
- Image Command
- Curve Command
- Touch control Command
- Work module Command

Etc.

*Command Set Table*

| Categories | Command | Command Parameter | Illustration |
|---|---|---|---|
| Handshake | 0x00 | No | Check the configuration and version |
| Parameter Configuration | 0x40 | Fcolor+Bcolor | Palette setting |
| | 0x41 | D_X(0x00-0x7F)+D_Y(0x00-0x7F) | Character space setting |
| | 0x42 | X+Y | Move the appointed color to background color palette |
| | 0x43 | X+Y | Move the appointed color to foreground color palette |
| | 0x44 | Mode+X+Y+Wide(0x01-0x1F)+Height(0x01-0x1F) | Cursor display mode setting |
| Text Display | 0x53 | X+Y+String | 8x8 lattice ASCII character |
| | 0x54 | | 16x16 lattice GBK |
| | 0x55 | | 32x32 GB2312 |
| | 0x6E | | 12X12 GBK |
| | 0x6F | | 24x24 GB2312 |
| | 0x98 | | Display and lattice, any encoded string |
| Points Setting | 0x50 | $(x,y)_0+(x,y)_1+\ldots\ldots+(x,y)_n$ | More points setting in the background color.(delete point) |
| | 0x51 | | More points in the foreground color. |
| | 0x74 | $X+Y_s+Y_e$ +Bcolor +$(y,Fcolor)_1+\ldots..+(y,Fcolor)_n$ | Dynamic curve display |
| | 0x72 | Address(H:M:L)+$Data_word_0$ +……$Data_word_n$ | Operation to the buffer of video card. |
| Line&Polygon | 0x56 | $(x,y)_0+(x,y)_1+\ldots\ldots+(x,y)_n$ | Polygon display:Line the points with foreground colored segment. |
| | 0x5D | | Polygon delete:Line the points with background colored segment |
| | 0x75 | $X+Y+Height_max+Height_0+Height_1$ +……$+Height_n$ | Spectrum display:display a continuous vertical line with the same end in a fast rhythm. |
| | 0x76 | $X+Y_dis(0x00-0xFF)+Y_0+Y_1+\ldots\ldots+Y_n$ | Line chat display(Xi=X+i*X_dis,Yi=Yi) |
| Acrs | 0x57 | $(Type,x,y,r)_0+(Type,x,y,r)_1+\ldots\ldots..+(Type,x,y,r)n$ | Arcs display |
| Rectangles | 0x59 | $(x_s,y_z,x_e,y_e)_0+(x_s,Y_z,x_e,y_e)_1+\ldots\ldots\ldots+(X_s,Y_z,X_e,Y_e)_n$ | Show rectangles:display rectangles by foreground color) |
| | 0x69 | | Delete rectangles:display rectangles by background color |
| Areas Operation | 0x64 | X+Y+Color | |
| | 0x52 | No | Clear the Screen |
| | 0x5A | $(x_s,y_z,xe,y_e)+(x_s,y_z,x_e,y_e)+\ldots\ldots+(x_s,y_z,x_e,y_e)_n$ | Areas Deleting |
| | 0x5B | | Fill in more than one appointed areas. |
| | 0x5C | | Areas color changing |
| | 0x60 | $(x_s,y_z,x_e,y_e,n)_0+(x_s,y_z,x_e,y_e,n)_1+\ldots\ldots+(x_s,y_z,x_e,y_e,n)_n$ | Appointed areas ring-shifting to the left |
| | 0x61 | | Appointed areas ring-shifting to the right |
| | 0x62 | | Appointed areas ring-shifting to the left |
| | 0x63 | | Appointed areas shifting to the right |
| Pictures&Icons | 0x70 | Picture_ID | Display a full screen image |
| | 0x7B | Picture_ID | Display a full screen image and calculate the cumulative sum. |
| | 0x71 | $Picture_ID+X_s+Y_s+X_e+Y_e+X+Y$ | Display part of a picture in the memory(background display) |
| | 0x9C | $Picture_ID+X_s+Y_s+Y_e+X_e+Y$ | Display a part from an image which stored in the module(background not shown),automatically restore the current image background . |
| | 0x9D | $Picture_ID+X_s+Y_s+X_e+Y_e+X+Y$ | Display part of a picture in the memory(background does not display) |
| | 0xE2 | Picture ID | Picture saving |
| | 0x99 | $(x,y,Icon_ID)_0+(x,y,Icon_ID)_1+\ldots\ldots+(x,y,Icon_ID)_n$ | User-defined icons display |
| Animation | 0x9A | 0Xff/Pack_ID | Turn off/on the automatic implementation of Command Set |
| Temporary | 0xC0 | Address(H:L)+$Data_word_0$+……..+$Data_word_n$ | Writing data to the temporary buffer |

| | | | |
|---|---|---|---|
| Buffer Operation | 0xC1 | 0x01+Address+Pixel_Number(H:L) | Display the pre-set date lines in the temporary buffer |
| | | 0x02+Address+Line_Number(H:L) | Display the pre-set date lines in the temporary buffer |
| | | 0x03+Address+X+Y+Line_Number+D_x+Dis_x+K_y+Color | Dynamic curve scaling:connecting the data points in the temporary buffer zone |
| | | 0x04+Addr1+X+Y+Line_Number+0x01+Dis_x+Color1+Addr0+Color0 | Oscillometer:connecting the data points in temporary buffer in a flicker-free high-speed |
| | | 0x05+Address+X+Y+Line_Number+D_x+Dis_x+M_y+D_y+Color | Using the data in the temporary buffer to display line charts. |
| | | 0x06+Address+X+Y+Line_Number+D_x+Dis_x+M_y+Color+Ymin+Ymax | Using the data in the temporary buffer zoom to display a window-constrained bi-directoional line chart |
| | | 0x10+Address+Frame_Number | Using the command in the temporary buffer to perform a synchronize display |
| | 0xC2 | <Address>+<Data_length> | Read back data from the temporary buffer. |
| Database Operation | 0xF2 | 0xF2+0xF2+0x5A+0xA5+Lib_ID | Font modifcation |
| | 0x90 | 0x55+0xAA+0x5A+0xA5+Address(H:MH:ML:L)+Data | Write data to the user's database(32MB) |
| | 0x91 | Address+Read_Length(H:L) | Read data from the database(32MB) |
| Keyboard Operation | 0x71 | K_code | Key code uploading |
| | 0xE5 | 0x55+0xAA+0x5A+0x+A5+K_Code$_0$+……+K_Code$_n$ | Key code port modification |
| Touchpad Operation | 0x72 | Touch_X+Touch_Y | Uploading the last data after the touch-screen is released,(which can turn off by 0xE0 Command) |
| | 0x73 | | Uploading data when pressing the touch panel(uploading once only by setting the command of 0xE0) |
| | 0xE4 | 0x55+0xAA+0x5A+0xA5 | Touch panel adjusting |
| | 0x78 | Touch Code | Uploading the defaulted key code when switching the touch interface. |
| | 0x79 | | |
| Buzzer Operation | 0x79 | BZ_time | Buzzing once only (10xBz_time mS) |
| Backlight Control | 0x5E | Non or 0x55+0x AA+0x5A+0xA5+V_ON+V_OFF+ON_TIME | Turn off the backlight or control the backlight mode by touching or keying |
| | 0x5F | Non or PWM_T(0x00-0x3F) | Turn the backlight on or adjusting the brightness by PWM. |
| Clock Operation | 0x 9B | 0x5A,0x5B(read)/0x00(off)/0xFF+M+TM+Color+X+Y(ON) | Clock on/off; read the clock |
| | 0x E7 | 0x55+0xAA+0x5A+0xA5+YY:MM:DD:HH:MM:SS | Clock adjusting |
| Parameter Configuration | 0x E0 | 0x55+0 x AA+0x5A+0x A5+Panel_Set+Bode_Set+Para1 | Configuring the user's serial port speed and the touch-screen data uploading. |
| Algorithm | 0xB0 | Download:0x01+PY_Code answer:0x01+HZ_num+String | |
| | | Download: 0x02+A+B+C+D answer:0x02+E+F | Calculating(AxB+C)/D,E is 4 bytes reminder |
| | | Download:0x03+Data_Pack0 answer:0x03+Data_Pack1 | Array listing of unsigned integers(2 bytes) |
| | | Download:0x04+PY_Code answer:0x04+HZ_num+String | PINYIN input based on GBK |
| Volume Operation | 0x30 | Start_Set+Play_number+Play_time | Play the music in the appointed zoom |
| | 0x32 | Volume_L+Volume_R+0x00 | Volume adjusting |
| | 0x33 | 0x55+0 x AA+0x5A | Stop playing |

# 8 Electrical Components

**This chapter contains the brands of the components:**

- TFT Panel
- Touch Screen
- CPU
- LCD Controller
- Flash memory
- Connecter
- Capacitance
- IC

| Components | Supplier |
|---|---|
| TFT Panel | INNOLUX 群創光電股份有限公司 AUO CHIMEI 世界级的液晶专家 SHARP |
| CPU | ST |
| LCD Controller | ALTERA MEASURABLE ADVANTAGE™ |
| Touch Screen | FUJITSU AMT Apex Material Technology Corp. |
| Flash Memory | SAMSUNG TURN ON TOMORROW |
| Connecter | molex one company › a world of innovation OMRON Sensing tomorrow™ Tyco Electronics |
| Capacitance | TDK muRata |
| IC | SAMSUNG TURN ON TOMORROW ISSI LINEAR TECHNOLOGY Texas Instruments |

# 9 Naming Rule

**This chapter contains the naming rule:**

**As sample STI070WT-01**

| Code | Explain |
|---|---|
| ST | Company Code |
| I | I=Industrial Type A=Advanced Type |
| 070 | TFT Panel Dimension: 7 inch |
| W | W=Wide Voltage (4.5V-20V) S=Super Wide Voltage (4.5V-42V) |
| T | T=With Resistive Touch Screen<br>N=Without Touch Screen<br>C=With Capacitive Touch Screen |
| 0 | 0=RS232 4=RS485 1=TTL |
| 1 | Hardware Code |

# 10 International Certification

**This chapter contains the certification we passed:**

- CE Certificate
- ROHS Certificate
- FCC Certificate
- ISO9001:2008 Quality System

## CE Certificate



TOBY

Shenzhen Toby Technology Co., Ltd.

10/F., A Block, Jiada R & D Bldg., No.5 Songpingshan Road, Science & Technology Park, Nanshan District, Shenzhen, China

CERTIFICATE OF CONFORMITY

No. : TB10098474

| | | |
|---|---|---|
| Applicant | : | Beijing Stone Technology Co.,Ltd |
| Address | : | 2807 Floor 28, Building B,Shiao International Center,Shaoyaoju North Road 101#,Chaoyang District,Beijing, China. |
| Manufacturer | : | Beijing Stone Technology Co.,Ltd |
| Address | : | 2807 Floor 28, Building B,Shiao International Center,Shaoyaoju North Road 101#,Chaoyang District,Beijing, China. |
| Product | : | Intellegent TFT-LCD Module |
| Models | : | STA070WT , STA080WT, STI070WT, STI080WT, eWin4300 eWinS070 , eWinI035 |

Test Standards:

EN 55022:2006+A1:2007;
EN 55024:1998+A1:2001+A2:2003.

The EUT described above has been tested by us with the listed standards and found in compliance with the Council Directive 2004/108/EC. It is possible to use CE marking to demonstrate the compliance with this EMC Directive.

The certificate applies to the tested sample above mentioned only and shall not imply an assessment of the whole production. It is only valid in connection with the test report number: TB-EMC108011.

CE

Justin Zhang
(Manager)
Sep. 21, 2010

## FCC Certificate



TOBY

Shenzhen Toby Technology Co., Ltd.

10/F., A Block, Jiada R & D Bldg.,No.5 Songpingshan Road, Science & Technology Park, Nanshan District, Shenzhen, China

VERIFICATION OF CONFORMITY

No.: TB10098475

| | | |
|---|---|---|
| Applicant | : | Beijing Stone Technology Co.,Ltd |
| Address | : | 2807 Floor 28, Building B,Shiao International Center,Shaoyaoju North Road 101#,Chaoyang District,Beijing, China. |
| Manufacturer | : | Beijing Stone Technology Co.,Ltd |
| Address | : | 2807 Floor 28, Building B,Shiao International Center,Shaoyaoju North Road 101#,Chaoyang District,Beijing, China. |
| Product | : | Intellegent TFT-LCD Module |
| Models | : | STA070WT , STA080WT, STI070WT, STI080WT, eWin4300 eWinS070 , eWinI035 |

Test Standard:

FCC Part 15: 2009 Subpart B

This device complies with Part 15 of the FCC rules, operation is subject to the following two conditions:
(1) This device may not cause harmful interference, and
(2) This device must accept any interference received, including interference that may cause undesired operation.

This is the results of test that was carried out by Shenzhen TOBY Technology Co., Ltd. From the submitted type samples of the product is conformity with the specification of the respective standards.
It is only valid in connection with the test report number: TB-FCC108012.

FC

Justin Zhang
(Manager)
Sep. 21, 2010

## RoHS Certificate



TOBY

Shenzhen Toby Technology Co., Ltd.

10/F., A Block, Jiada R & D Bldg.,No.5 Songpingshan Road, Science & Technology Park, Nanshan District, Shenzhen, China

VERIFICATION OF CONFORMITY

No.: TB10098475

| | | |
|---|---|---|
| Applicant | : | Beijing Stone Technology Co.,Ltd |
| Address | : | 2807 Floor 28, Building B,Shiao International Center,Shaoyaoju North Road 101#,Chaoyang District,Beijing, China. |
| Manufacturer | : | Beijing Stone Technology Co.,Ltd |
| Address | : | 2807 Floor 28, Building B,Shiao International Center,Shaoyaoju North Road 101#,Chaoyang District,Beijing, China. |
| Product | : | Intellegent TFT-LCD Module |
| Models | : | STA070WT , STA080WT, STI070WT, STI080WT, eWin4300 eWinS070 , eWinI035 |

Test Standard:

Part 15: 2009 Subpart B

This device complies with Part 15 of the FCC rules, operation is subject to the following two conditions:
(1) This device may not cause harmful interference, and
(2) This device must accept any interference received, including interference that may cause undesired operation.

This is the results of test that was carried out by Shenzhen TOBY Technology Co., Ltd. From the submitted type samples of the product is conformity with the specification of the respective standards.
It is only valid in connection with the test report number: TB-FCC108012.

RoHS

Justin Zhang
(Manager)
Sep. 21, 2010

## ISO9001:2008



中经认证

No: 04408Q10576R0S

QUALITY MANAGEMENT SYSTEM CERTIFICATION

*Certificate*

This is to certify the quality management system of

BEIJING STONE TECHNOLOGY CO., LTD.

REGISTER ADD/ 1ST AUDIT ADD: ROOM 901-904, HAOJING MANSION, BUILDING 1, NO. 108, ZHICHUN ROAD, HAIDIAN DISTRICT, BEIJING
2ND AUDIT ADD: BASEMENT, BUILDING 1, QUNYING SCIENCE & TECHNOLOGY PARK, NO.8, CHUANGYE ROAD, SHANGDI INFORMATION INDUSTRY BASE, HAIDIAN DISTRICT, BEIJING

is in conformity with

GB/T19001-2008 idt ISO9001 : 2008 standard

The certificate is valid for the following scope:

DESIGN, MANUFACTURE AND SERVICE OF SMART LCD DISPLAY TERMINAL

Issue Date: 2008-06-17 Expiry Date: 2011-06-16 Alteration Date: 2010-07-21

Within the certificate validity , the organization must pass the annual surveillance audit and get the mark before the prescribed date. The certificate is valid ONLY if the mark is present since the first surveillance audit.

Beijing Zhongjing Quality Certification Co., Ltd. General Manager

Date: 2010-07-21

CNAS
MANAGEMENT SYSTEM
CNAS C044- Q

IAF

# APPENDIX

**A MCU Sample Program**

**B MCU Circuit Design**

**C ESD Guidelines**

# A. MCU Sample Program

```
C8051 MCU C Language
//-----------------------------------------------------------------------------
// Includes
//-----------------------------------------------------------------------------
#include<reg52.h>

//-----------------------------------------------------------------------------
// sbit  Definitions
//-----------------------------------------------------------------------------
sbit  LED=P0^0;

//-----------------------------------------------------------------------------
// Global CONSTANTS
//-----------------------------------------------------------------------------
#define SYSCLK      22118400                    // “SYSCLK frequency in Hz”
#define BAUD_RATE   115200                      // “Baud rate”
#define uchar unsigned char
#define uint unsigned int
uchar pic[3]={0xAA,0x70,0x08};
/-----------------------------------------------------------------------------
// Function PROTOTYPES
//-----------------------------------------------------------------------------
void Uart0_transmit(uchar i);                   // “Send a byte to the terminal”
void send_str(uchar *p,uchar s);                // “Send a string to the terminal”
void delay_ms(uchar n);                         // “Delay”
void SysInit(void);                             // “Initialization of system”
void en(void);                                  // “Frame end”
void pic_str(uchar i);                          // “Picture switching sub-function”

//-----------------------------------------------------------------------------
// Uart0_transmit
//-----------------------------------------------------------------------------
void Uart0_transmit(uchar i)                    // “Send 1 byte to terminal”
{
                        ES=0;
                        TI=0;
                        SBUF=i;                 // “Send data to uart0”
                        while (!TI);            // “Wait for the finish of sending a byte”
                        TI=                     // “Clear the interruption mark”
                        ES=1;
}
void send_str(uchar *p,uchar s)                 // “Send a string to the terminal”
{
                        uchar m;
                        for(m=0;m<s;m++)
                        {
                           Uart0_transmit(*p);
                           p++;
                        }
}
//-----------------------------------------------------------------------------
// delay                                        // “Delay sub-function”
//-----------------------------------------------------------------------------
```

```
void delay_ms(uchar n)
{
                        uint i,j;
                        for(i=1000;i>0;i--) {
                        for(j=25*n;j>0;j--) {;}
                        }
}

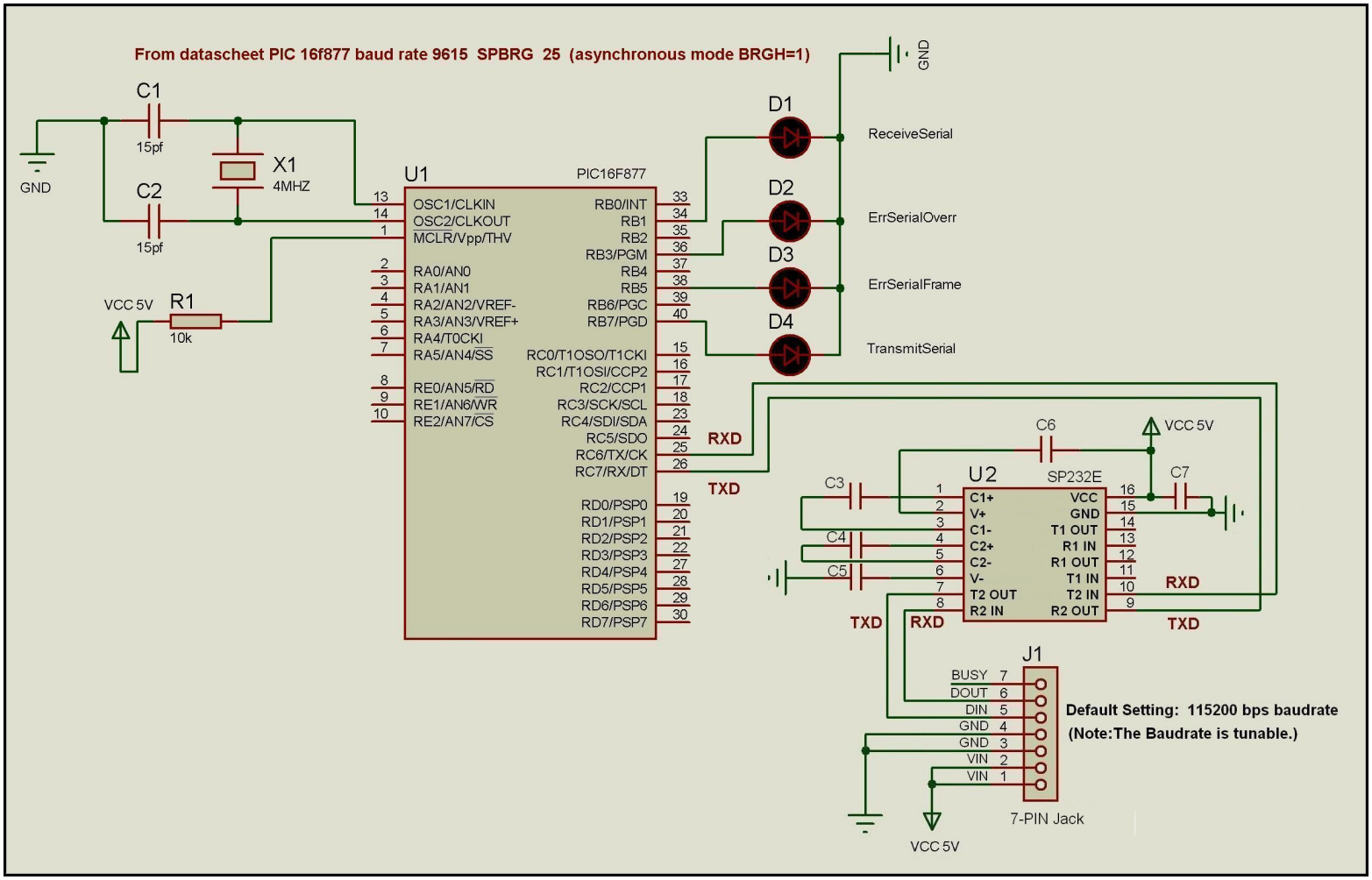
//--------------------------------------------------------------------------
// SysInit                                          // “Initialization of system”
//--------------------------------------------------------------------------
void SysInit(void)
{
                                PCON |=0x80;
                                SCON=0x50;
                                TMOD=0x21;
                                TH1=255;
                                TL1=255;
                                TR1=1;
            ES=0;
            TH0=0xDC;
            TL0=0x00;
            TR0=1;
            ET0=1;
}

//--------------------------------------------------------------------------
// pic_str                                          // “Picture switching sub-function”
//--------------------------------------------------------------------------
void pic_str(uchar i)
{
                pic[2]=i;
                send_str(pic,3);          // “Send the command of picture switching”
                en();
}

//--------------------------------------------------------------------------
// main() Routine
//--------------------------------------------------------------------------
void main (void)                                  // “main function”
{
  EA=0;                                           // “Close Interruption”
  SysInit();
  EA=1;                                           // “Open Interruption”
  delay_ms(40);

    while (1)
          {
      pic_str();                                  // “Picture switching”
          }
           Return 0;
}
//--------------------------------------------------------------------------
// End Of File
//--------------------------------------------------------------------------
```

# B. MCU Circuit Design



# C. ESD Guidelines

**What does ESD mean?**

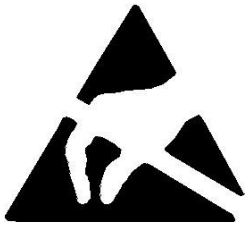
Virtually all present-day modules incorporate highly integrated MOS devices or components. For technological reasons, these electronic components are very sensitive to overvoltages and consequently therefore to electrostatic discharge:

These devices are referred to in German as Elektrostatisch Gefährdeten Bauelemente/ Baugruppen: ºEGBº

The more frequent international name is:

ºESDº (Electrostatic Sensitive Device)

The following symbol on plates on cabinets, mounting racks or packages draws attention to the use of electrostatic sensitive devices and thus to the contact sensitivity of the assemblies concerned:



**ESDs** may be destroyed by voltages and energies well below the perception threshold of persons. Voltages of this kind occur as soon as a device or an assembly is touched by a person who is not electrostatically discharged . Devices exposed to such overvoltages cannot immediately be detected as defective in the majority of cases since faulty behavior may occur only after a long period of operation.

**Precautions against electrostatic discharge**

Most plastics are capable of carrying high charges and it is therefore imperative that they be kept away from sensitive components.

When handling electrostatic sensitive devices, make sure that persons, workplaces and packages are properly grounded.

**Handling ESD assemblies**

A general rule is that assemblies should be touched only when this cannot be avoided owing to the work that has to performed on them. Under no circumstances should you handle printed-circuit boards by touching device pins or circuitry.

You should touch devices only if

- you are grounded by permanently wearing an ESD wrist strap or
- you are wearing ESD shoes or ESD shoe-grounding protection straps in conjunction with an ESD floor.

Before you touch an electronic assembly, your body must be discharged. The simplest way of doing this is to touch a conductive, grounded object immediately beforehand ± for example, bare metal parts of a cabinet, water pipe etc.

Assemblies should not be brought into contact with charge-susceptible and highly insulating materials such as plastic films, insulating table tops and items of clothing etc. containing synthetic fibers.

Assemblies should be deposited only on conductive surfaces (tables with an ESD coating, conductive ESD cellular material, ESD bags, ESD shipping containers).

Do not place assemblies near visual display units, monitors or television sets (minimum distance to screen > 10 cm).

**Measuring and modifying ESD assemblies**

Perform measurements on ESD assemblies only when

- the measuring instrument is grounded ± for example, by means of a protective conductor ± or
- the measuring head has been briefly discharged before measurements are made with a potential-free measuring instrument ± for example, by touching a bare metal control cabinet.

When soldering, use only grounded soldering irons.

**Shipping ESD assemblies**

Always store and ship assemblies and devices in conductive packing ± for example, metallized plastic boxes and tin cans.

If packing is not conductive, assemblies must be conductively wrapped before they are packed. You can use, for example, conductive foam rubber, ESD bags, domestic aluminum foil or paper (never use plastic bags or foils).

With assemblies containing fitted batteries, make sure that the conductive packing does not come into contact with or short-circuit battery connectors. If necessary, cover the connectors beforehand with insulating tape or insulating material.

# Glossary



**Baud rate**

Rate of speed at which data is downloaded. Baud rate is specified in Bit/s.

**Boot**

A loading process which downloads the operating system in the working memory of the operating unit.



**Command Set**

Hex Code, the MCU can control the TFT Module via the command set.

**Configuration file**

It can be created by the softwares.



**Download**

Download the image, configuration files and data through mini USB port or SD card.

**Download mode**

Through mini USB port or SD card.



**Flash memory**

Programmable memory which can be electrically deleted and written to again segment-by-segment.



**Half Brightness Life**

The period of time after which the brightness tube only achieves 50% of the original value.



**Input field**

Enables the user to enter values which are subsequently send to the **MCU**.



**MCU**

Micro Control Unit, it is widely used in the industrial control.



**Normal operation**

Operating unit operating mode in which messages are displayed and screens can be operated.



**Output field**

Displays current values from the **MCU** on the operating unit.



**Process screen**

The display of process values and process progress on the operating unit in the form of screens, which may contain graphics, texts and values.



**RS485**

Standard interface for serial data transfer at a very high transmission rate.



**Screen**

A screen displays all the logically related process data on the operating unit, whereby the individual values can be modified.



**Touch panel**

This is an operating unit without a keyboard. The touch panel (abbreviated to TP) is operate via the contact-sensitive screen elements.